

- 1 (a) (i) Magnetic flux density B generated by a magnet may be found by determining the magnetic flux ϕ passing normally through an area A according to the expression

$$\phi = BA$$

In one experiment, ϕ and A were determined to be $(50 \pm 5) \mu\text{Wb}$ and $(4.00 \pm 0.01) \times 10^{-4} \text{ m}^2$ respectively.

Determine the value of B along with its associated uncertainty.

$$B = \dots\dots\dots \pm \dots\dots\dots \text{ T [3]}$$

- (ii) Explain why the value of B cannot be more precisely determined even if the precision in the instrument used to measure A is improved.

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 [1]

- (b) A non-uniform bar AB makes an angle of 60° with a horizontal surface, as shown in Fig. 1.1.

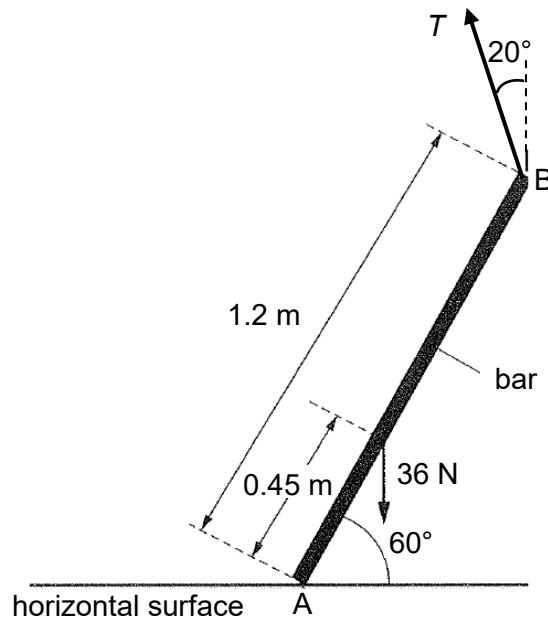


Fig. 1.1

The bar is hinged at A and is supported by a light string at B. The string is inclined at an angle of 20° to the vertical. The bar has a length of 1.2 m and a weight of 36 N. The centre of gravity of the bar is 0.45 m from A.

(i) Show that the magnitude of the tension T in the string is 8.8 N.

[2]

(ii) A force F acts on the bar at A.

Calculate the magnitude of F .

$F = \dots\dots\dots$ N [3]

2 As part of a physics lesson demonstration, a student stands between a sound speaker and